VK-RCD First-Round HPE and HAR Experiment Report

Auto-generated Result Summary

May 7, 2026

Abstract

This report summarizes the first-round VK-RCD experiments for HPE and HAR. It includes completion auditing, official X-Fi comparisons, HPE robustness analysis, and a result-file index. Blue values indicate improvements over official X-Fi, while red values indicate degradation. HPE uses MPJPE and PA-MPJPE, where lower is better. HAR uses Accuracy and Macro-F1, where higher Accuracy is better.

Contents

1 Completion Audit

This section reports the first-round local result inventory. DONE means that the expected checkpoint or evaluation CSV exists locally. MISSING entries are not fabricated and must be completed or resynchronized before final paper submission.

Table 1: Current HPE and HAR completion summary

Task	DONE	MISSING_MODEL	MISSING_EVAL
HPE	8	18	20
HAR	24	1	0

Table 2: Missing result items

Task	Type	Item	Status	Expected path
HPE	model	ablation/no_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/output_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/output_token_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/output_bone_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/full_structural_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/uniform/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/attention/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/uncertainty/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/mlp_vk/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	ablation/skeleton_prompt/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\
HPE	model	teacher_full_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\teacher_fu
HPE	model	student_vk_missing_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_v
HPE	model	student_nv_missing_cross_scene/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_n
HPE	model	baseline_full_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\baseline_fu
HPE	model	teacher_full_cross_subject_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\teacher_fu

Task	Type	Item	Status	Expected path
HPE	model	student_vk_missing_cross_subject_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_vl
HPE	model	student_nv_missing_cross_subject/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_nv
HPE	model	baseline_full_cross_subject_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\baseline_fu
HPE	eval	student_nv_nonvisual_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\stude
HPE	eval	baseline_full_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\basel
HPE	eval	ablation_distillation_no_kd_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_distillation_output_kd_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_distillation_output_token_kd_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_distillation_output_bone_kd_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_distillation_full_structural_kd_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_reliability_uniform_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_reliability_attention_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_reliability_uncertainty_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_encoder_mlp_vk_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	ablation_encoder_skeleton_prompt_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablat
HPE	eval	teacher_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\teach
HPE	eval	student_vk_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\stude
HPE	eval	student_nv_cross_scene_nonvisual_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\stude
HPE	eval	baseline_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\basel
HPE	eval	teacher_cross_subject_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\teach
HPE	eval	student_vk_cross_subject_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\stude
HPE	eval	student_nv_cross_subject_nonvisual_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\stude
HPE	eval	baseline_cross_subject_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\basel
HAR	model	student_vk_missing_cross_subject/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HAR\outputs\student_vl

2 HPE Results

The HPE comparison contains 62 comparable rows. MPJPE improves in 34 rows, and PA-MPJPE improves in 33 rows. All HPE errors are reported in millimeters, and lower is better.

Important note on the Teacher. The full-modality Teacher is not a missing-modality model. It is trained as a full-modality oracle with all modalities available, so evaluating it under single-modality or arbitrary missing-modality settings creates a distribution shift. Therefore, the Teacher may look very poor in many partial-modality rows, while its full-modality row is strong. The Student-VK and Student-NV models are the models intended for missing-modality robustness.

Table 3: HPE full-modality comparison against official X-Fi

Method	Source	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ PA-MPJPE
VK-RCD-Teacher	teacher_all_combinations.csv	48.45	83.70	-35.25	-14.54
VK-RCD-Student-VK	all_combinations.csv	50.17	83.70	-33.53	-14.12

Table 4: HPE per-modality comparison against official X-Fi

Method	Split	Modality	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ PA
VK-RCD-Teacher	random_split	vk	487.99	93.90	+394.09	+79.05
VK-RCD-Teacher	random_split	depth	234.48	101.80	+132.68	+36.20
VK-RCD-Teacher	random_split	lidar	1615.42	167.10	+1448.32	+284.13
VK-RCD-Teacher	random_split	mmwave	4196.64	127.40	+4069.24	+228.92
VK-RCD-Teacher	random_split	wifi-csi	4173.33	225.60	+3947.73	+272.92
VK-RCD-Teacher	random_split	vk+depth	369.12	86.10	+283.02	+49.21
VK-RCD-Teacher	random_split	vk+lidar	1072.74	93.00	+979.74	+153.37
VK-RCD-Teacher	random_split	vk+mmwave	814.71	88.80	+725.91	+116.75
VK-RCD-Teacher	random_split	vk+wifi-csi	2491.83	93.00	+2398.83	+275.70
VK-RCD-Teacher	random_split	depth+lidar	285.80	102.50	+183.30	+39.07
VK-RCD-Teacher	random_split	depth+mmwave	178.87	98.00	+80.87	+18.26

Method	Split	Modality	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ PA
VK-RCD-Teacher	random_split	depth+wifi-csi	197.61	101.80	+95.81	+25.97
VK-RCD-Teacher	random_split	lidar+mmwave	2157.14	109.80	+2047.34	+324.25
VK-RCD-Teacher	random_split	lidar+wifi-csi	4007.08	159.50	+3847.58	+281.43
VK-RCD-Teacher	random_split	mmwave+wifi-csi	4675.91	117.20	+4558.71	+210.45
VK-RCD-Teacher	random_split	vk+depth+lidar	634.60	84.80	+549.80	+64.07
VK-RCD-Teacher	random_split	vk+depth+mmwave	465.16	83.40	+381.76	+39.15
VK-RCD-Teacher	random_split	vk+depth+wifi-csi	100.71	85.30	+15.41	+3.37
VK-RCD-Teacher	random_split	vk+lidar+mmwave	1035.05	88.40	+946.65	+180.75
VK-RCD-Teacher	random_split	vk+lidar+wifi-csi	4736.11	93.00	+4643.11	+211.53
VK-RCD-Teacher	random_split	vk+mmwave+wifi-csi	4769.71	88.50	+4681.21	+198.88
VK-RCD-Teacher	random_split	depth+lidar+mmwave	310.49	96.00	+214.49	+28.62
VK-RCD-Teacher	random_split	depth+lidar+wifi-csi	177.40	102.00	+75.40	+18.88
VK-RCD-Teacher	random_split	depth+mmwave+wifi-csi	121.88	97.00	+24.88	+7.73
VK-RCD-Teacher	random_split	lidar+mmwave+wifi-csi	4360.45	107.40	+4253.05	+259.17
VK-RCD-Teacher	random_split	vk+depth+lidar+mmwave	593.34	83.50	+509.84	+45.64
VK-RCD-Teacher	random_split	vk+depth+lidar+wifi-csi	67.89	86.00	-18.11	-7.67
VK-RCD-Teacher	random_split	vk+depth+mmwave+wifi-csi	71.27	84.00	-12.73	-8.69
VK-RCD-Teacher	random_split	vk+lidar+mmwave+wifi-csi	4876.08	88.60	+4787.48	+182.08
VK-RCD-Teacher	random_split	depth+lidar+mmwave+wifi-csi	112.99	97.60	+15.39	+3.32
VK-RCD-Teacher	random_split	vk+depth+lidar+mmwave+wifi-csi	48.45	83.70	-35.25	-14.54
VK-RCD-Student-VK	random_split	vk	92.66	93.90	-1.24	-17.70
VK-RCD-Student-VK	random_split	depth	53.88	101.80	-47.92	-12.79
VK-RCD-Student-VK	random_split	lidar	139.90	167.10	-27.20	-9.76
VK-RCD-Student-VK	random_split	mmwave	114.98	127.40	-12.42	-11.67
VK-RCD-Student-VK	random_split	wifi-csi	222.14	225.60	-3.46	+4.89
VK-RCD-Student-VK	random_split	vk+depth	51.65	86.10	-34.45	-13.81
VK-RCD-Student-VK	random_split	vk+lidar	73.18	93.00	-19.82	-17.78
VK-RCD-Student-VK	random_split	vk+mmwave	68.93	88.80	-19.87	-18.40
VK-RCD-Student-VK	random_split	vk+wifi-csi	88.11	93.00	-4.89	-17.20
VK-RCD-Student-VK	random_split	depth+lidar	53.00	102.50	-49.50	-13.05
VK-RCD-Student-VK	random_split	depth+mmwave	52.32	98.00	-45.68	-12.66
VK-RCD-Student-VK	random_split	depth+wifi-csi	53.54	101.80	-48.26	-12.58
VK-RCD-Student-VK	random_split	lidar+mmwave	97.54	109.80	-12.26	-8.31
VK-RCD-Student-VK	random_split	lidar+wifi-csi	134.11	159.50	-25.39	-9.48
VK-RCD-Student-VK	random_split	mmwave+wifi-csi	108.54	117.20	-8.66	-5.43
VK-RCD-Student-VK	random_split	vk+depth+lidar	51.10	84.80	-33.70	-14.04
VK-RCD-Student-VK	random_split	vk+depth+mmwave	50.50	83.40	-32.90	-13.78
VK-RCD-Student-VK	random_split	vk+depth+wifi-csi	51.63	85.30	-33.67	-13.81
VK-RCD-Student-VK	random_split	vk+lidar+mmwave	66.08	88.40	-22.32	-18.43
VK-RCD-Student-VK	random_split	vk+lidar+wifi-csi	70.68	93.00	-22.32	-17.94
VK-RCD-Student-VK	random_split	vk+mmwave+wifi-csi	66.69	88.50	-21.81	-18.30
VK-RCD-Student-VK	random_split	depth+lidar+mmwave	51.92	96.00	-44.08	-12.80
VK-RCD-Student-VK	random_split	depth+lidar+wifi-csi	52.86	102.00	-49.14	-12.72
VK-RCD-Student-VK	random_split	depth+mmwave+wifi-csi	52.13	97.00	-44.87	-12.48
VK-RCD-Student-VK	random_split	lidar+mmwave+wifi-csi	94.20	107.40	-13.20	-8.00
VK-RCD-Student-VK	random_split	vk+depth+lidar+mmwave	50.25	83.50	-33.25	-14.15
VK-RCD-Student-VK	random_split	vk+depth+lidar+wifi-csi	51.05	86.00	-34.95	-14.02

Method	Split	Modality	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ PA
VK-RCD-Student-VK	random_split	vk+depth+mmwave+wifi-csi	50.41	84.00	-33.59	-14.06
VK-RCD-Student-VK	random_split	vk+lidar+mmwave+wifi-csi	64.22	88.60	-24.38	-18.35
VK-RCD-Student-VK	random_split	depth+lidar+mmwave+wifi-csi	51.78	97.60	-45.82	-12.86
VK-RCD-Student-VK	random_split	vk+depth+lidar+mmwave+wifi-csi	50.17	83.70	-33.53	-14.12

2.1 VK Noise and Dropped-Joint Robustness

Table 5: HPE Student-VK robustness under VK noise and joint dropping

Source	Noise Std	Drop Prob	MPJPE	PA-MPJPE	MSE	
student_vk_noise_robustness.csv	0.0		50.18	33.48	0.0014590788163282586	VK-RCD-St
student_vk_noise_robustness.csv	0.0		52.64	34.57	0.0015710848316534925	VK-RCD-St
student_vk_noise_robustness.csv	0.0		54.10	35.19	0.0016379341990311823	VK-RCD-St
student_vk_noise_robustness.csv	0.0		54.75	35.50	0.0016690452336497448	VK-RCD-St
student_vk_noise_robustness.csv	5.0		50.88	33.92	0.001488102523395341	VK-RCD-St
student_vk_noise_robustness.csv	5.0		52.88	34.71	0.001581960488836217	VK-RCD-St
student_vk_noise_robustness.csv	5.0		54.12	35.19	0.0016401267167432682	VK-RCD-St
student_vk_noise_robustness.csv	5.0		54.76	35.50	0.0016697228733739478	VK-RCD-St
student_vk_noise_robustness.csv	10.0		52.08	34.57	0.001539294305564543	VK-RCD-St
student_vk_noise_robustness.csv	10.0		53.26	34.93	0.0015987949889641724	VK-RCD-St
student_vk_noise_robustness.csv	10.0		54.12	35.20	0.001638386860237358	VK-RCD-St
student_vk_noise_robustness.csv	10.0		54.74	35.52	0.0016693329612529602	VK-RCD-St
student_vk_noise_robustness.csv	20.0		53.63	35.20	0.001611003164998953	VK-RCD-St
student_vk_noise_robustness.csv	20.0		53.85	35.15	0.001621275579426224	VK-RCD-St
student_vk_noise_robustness.csv	20.0		54.21	35.25	0.0016435792571283744	VK-RCD-St
student_vk_noise_robustness.csv	20.0		54.78	35.54	0.0016701842608368845	VK-RCD-St

3 HAR Results

The HAR comparison contains 75 comparable rows. Accuracy improves over official X-Fi in 50 rows. Accuracy and Macro-F1 are reported as percentages, and higher accuracy is better.

Table 6: HAR full-modality comparison against official X-Fi

Method	Source	Ours Acc.	X-Fi Acc.	Δ Acc.
VK-RCD-HAR-Teacher	teacher_random_all_combinations.csv	95.57	72.20	+23.37
VK-RCD-HAR-Student-VK-Missing	student_vk_random_all_combinations.csv	96.56	72.20	+24.36
VK-RCD-HAR-Teacher	all_combinations.csv	95.59	72.20	+23.39
VK-RCD-HAR-Ablation-NoKD	ablation_no_distill_all_combinations.csv	95.99	72.20	+23.79
VK-RCD-HAR-Ablation-UniformFusion	ablation_uniform_fusion_all_combinations.csv	95.39	72.20	+23.19

Table 7: HAR per-modality comparison against official X-Fi

Method	Split	Modality	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Teacher	random_split	vk	3.40	26.50	-23.10	0.45
VK-RCD-HAR-Teacher	random_split	depth	83.71	48.10	+35.61	82.80
VK-RCD-HAR-Teacher	random_split	lidar	3.36	52.70	-49.34	1.17
VK-RCD-HAR-Teacher	random_split	mmwave	66.72	85.70	-18.98	64.78
VK-RCD-HAR-Teacher	random_split	vk+depth	83.50	45.30	+38.20	82.60

Method	Split	Modality	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Teacher	random_split	vk+lidar	3.91	35.20	-31.29	1.31
VK-RCD-HAR-Teacher	random_split	vk+mmwave	66.46	73.40	-6.94	63.67
VK-RCD-HAR-Teacher	random_split	depth+lidar	83.37	51.60	+31.77	82.54
VK-RCD-HAR-Teacher	random_split	depth+mmwave	95.76	79.80	+15.96	95.74
VK-RCD-HAR-Teacher	random_split	lidar+mmwave	66.69	88.70	-22.01	63.96
VK-RCD-HAR-Teacher	random_split	vk+depth+lidar	83.66	48.70	+34.96	82.91
VK-RCD-HAR-Teacher	random_split	vk+depth+mmwave	95.66	70.70	+24.96	95.65
VK-RCD-HAR-Teacher	random_split	vk+lidar+mmwave	66.39	77.80	-11.41	63.36
VK-RCD-HAR-Teacher	random_split	depth+lidar+mmwave	95.64	80.50	+15.14	95.63
VK-RCD-HAR-Teacher	random_split	vk+depth+lidar+mmwave	95.57	72.20	+23.37	95.56
VK-RCD-HAR-Student-VK-Missing	random_split	vk	65.31	26.50	+38.81	65.70
VK-RCD-HAR-Student-VK-Missing	random_split	depth	88.02	48.10	+39.92	87.86
VK-RCD-HAR-Student-VK-Missing	random_split	lidar	40.80	52.70	-11.90	40.10
VK-RCD-HAR-Student-VK-Missing	random_split	mmwave	85.88	85.70	+0.18	86.06
VK-RCD-HAR-Student-VK-Missing	random_split	vk+depth	89.51	45.30	+44.21	89.36
VK-RCD-HAR-Student-VK-Missing	random_split	vk+lidar	68.54	35.20	+33.34	69.05
VK-RCD-HAR-Student-VK-Missing	random_split	vk+mmwave	92.75	73.40	+19.35	92.82
VK-RCD-HAR-Student-VK-Missing	random_split	depth+lidar	88.52	51.60	+36.92	88.39
VK-RCD-HAR-Student-VK-Missing	random_split	depth+mmwave	96.45	79.80	+16.65	96.44
VK-RCD-HAR-Student-VK-Missing	random_split	lidar+mmwave	89.51	88.70	+0.81	89.57
VK-RCD-HAR-Student-VK-Missing	random_split	vk+depth+lidar	89.62	48.70	+40.92	89.49
VK-RCD-HAR-Student-VK-Missing	random_split	vk+depth+mmwave	96.56	70.70	+25.86	96.55
VK-RCD-HAR-Student-VK-Missing	random_split	vk+lidar+mmwave	92.73	77.80	+14.93	92.81
VK-RCD-HAR-Student-VK-Missing	random_split	depth+lidar+mmwave	96.44	80.50	+15.94	96.43
VK-RCD-HAR-Student-VK-Missing	random_split	vk+depth+lidar+mmwave	96.56	72.20	+24.36	96.55
VK-RCD-HAR-Teacher	random_split	vk	3.40	26.50	-23.10	0.45
VK-RCD-HAR-Teacher	random_split	depth	83.71	48.10	+35.61	82.80
VK-RCD-HAR-Teacher	random_split	lidar	3.45	52.70	-49.25	1.22
VK-RCD-HAR-Teacher	random_split	mmwave	66.72	85.70	-18.98	64.78
VK-RCD-HAR-Teacher	random_split	vk+depth	83.50	45.30	+38.20	82.60
VK-RCD-HAR-Teacher	random_split	vk+lidar	3.85	35.20	-31.35	1.26
VK-RCD-HAR-Teacher	random_split	vk+mmwave	66.46	73.40	-6.94	63.67
VK-RCD-HAR-Teacher	random_split	depth+lidar	83.39	51.60	+31.79	82.57
VK-RCD-HAR-Teacher	random_split	depth+mmwave	95.76	79.80	+15.96	95.74
VK-RCD-HAR-Teacher	random_split	lidar+mmwave	66.64	88.70	-22.06	63.92
VK-RCD-HAR-Teacher	random_split	vk+depth+lidar	83.70	48.70	+35.00	82.97
VK-RCD-HAR-Teacher	random_split	vk+depth+mmwave	95.66	70.70	+24.96	95.65
VK-RCD-HAR-Teacher	random_split	vk+lidar+mmwave	66.33	77.80	-11.47	63.31
VK-RCD-HAR-Teacher	random_split	depth+lidar+mmwave	95.62	80.50	+15.12	95.61
VK-RCD-HAR-Teacher	random_split	vk+depth+lidar+mmwave	95.59	72.20	+23.39	95.58
VK-RCD-HAR-Ablation-NoKD	random_split	vk	60.94	26.50	+34.44	59.31
VK-RCD-HAR-Ablation-NoKD	random_split	depth	86.85	48.10	+38.75	86.78

Method	Split	Modality	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Ablation-NoKD	random_split	lidar	33.14	52.70	-19.56	31.11
VK-RCD-HAR-Ablation-NoKD	random_split	mmwave	83.13	85.70	-2.57	83.31
VK-RCD-HAR-Ablation-NoKD	random_split	vk+depth	88.09	45.30	+42.79	87.94
VK-RCD-HAR-Ablation-NoKD	random_split	vk+lidar	63.63	35.20	+28.43	62.65
VK-RCD-HAR-Ablation-NoKD	random_split	vk+mmwave	91.05	73.40	+17.65	91.12
VK-RCD-HAR-Ablation-NoKD	random_split	depth+lidar	87.24	51.60	+35.64	87.25
VK-RCD-HAR-Ablation-NoKD	random_split	depth+mmwave	95.57	79.80	+15.77	95.56
VK-RCD-HAR-Ablation-NoKD	random_split	lidar+mmwave	85.92	88.70	-2.78	86.09
VK-RCD-HAR-Ablation-NoKD	random_split	vk+depth+lidar	88.62	48.70	+39.92	88.53
VK-RCD-HAR-Ablation-NoKD	random_split	vk+depth+mmwave	95.96	70.70	+25.26	95.95
VK-RCD-HAR-Ablation-NoKD	random_split	vk+lidar+mmwave	91.23	77.80	+13.43	91.29
VK-RCD-HAR-Ablation-NoKD	random_split	depth+lidar+mmwave	95.50	80.50	+15.00	95.50
VK-RCD-HAR-Ablation-NoKD	random_split	vk+depth+lidar+mmwave	95.99	72.20	+23.79	95.98
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk	5.52	26.50	-20.98	1.32
VK-RCD-HAR-Ablation-UniformFusion	random_split	depth	85.59	48.10	+37.49	85.30
VK-RCD-HAR-Ablation-UniformFusion	random_split	lidar	2.65	52.70	-50.05	1.55
VK-RCD-HAR-Ablation-UniformFusion	random_split	mmwave	69.91	85.70	-15.79	68.71
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+depth	86.48	45.30	+41.18	86.24
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+lidar	4.58	35.20	-30.62	1.76
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+mmwave	67.75	73.40	-5.65	66.21
VK-RCD-HAR-Ablation-UniformFusion	random_split	depth+lidar	86.16	51.60	+34.56	85.91
VK-RCD-HAR-Ablation-UniformFusion	random_split	depth+mmwave	95.54	79.80	+15.74	95.54
VK-RCD-HAR-Ablation-UniformFusion	random_split	lidar+mmwave	67.80	88.70	-20.90	66.56
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+depth+lidar	86.43	48.70	+37.73	86.19
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+depth+mmwave	95.50	70.70	+24.80	95.49
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+lidar+mmwave	67.09	77.80	-10.71	65.08
VK-RCD-HAR-Ablation-UniformFusion	random_split	depth+lidar+mmwave	95.42	80.50	+14.92	95.42
VK-RCD-HAR-Ablation-UniformFusion	random_split	vk+depth+lidar+mmwave	95.39	72.20	+23.19	95.39

4 Result File Index

The report is generated from the following CSV files.

- reports/completion_audit.csv
- reports/HPE/hpe_official_comparison.csv
- reports/HPE/hpe_robustness_results.csv
- reports/HAR/har_official_comparison.csv
- HPE/legacy_xfi/xfi_hpe_table1.csv
- HAR/legacy_xfi/xfi_har_table6.csv